

MBAi 448 | Winter 2026

Agentic AI

Attendance

Enter the “Magic Word” in Canvas

Do Not Share with Absent Students





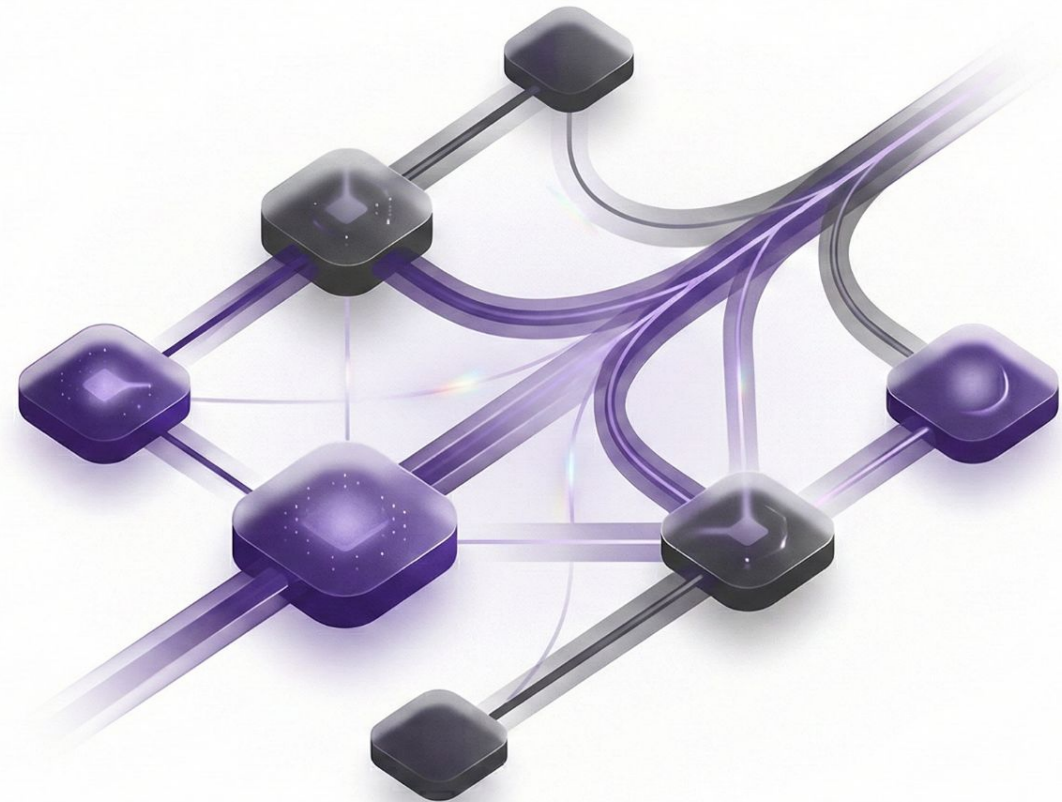
REMINDER

Logos DUE by end of this week!

Question

Have you seen actual AI agents used in the real world?

Agents & Agentic Workflows



Autonomous AI systems that plan, act, and adapt across multi-step tasks.

Autonomously planning, using tools, using memory, and executing complex workflows to achieve goals.

From Chatbots to Agents

- Chatbots:
 - Single-turn: prompt-in › response-out
- Agents:
 - Autonomous
 - Multi-step
 - Tool use
 - Memory
- Shift from interaction to execution

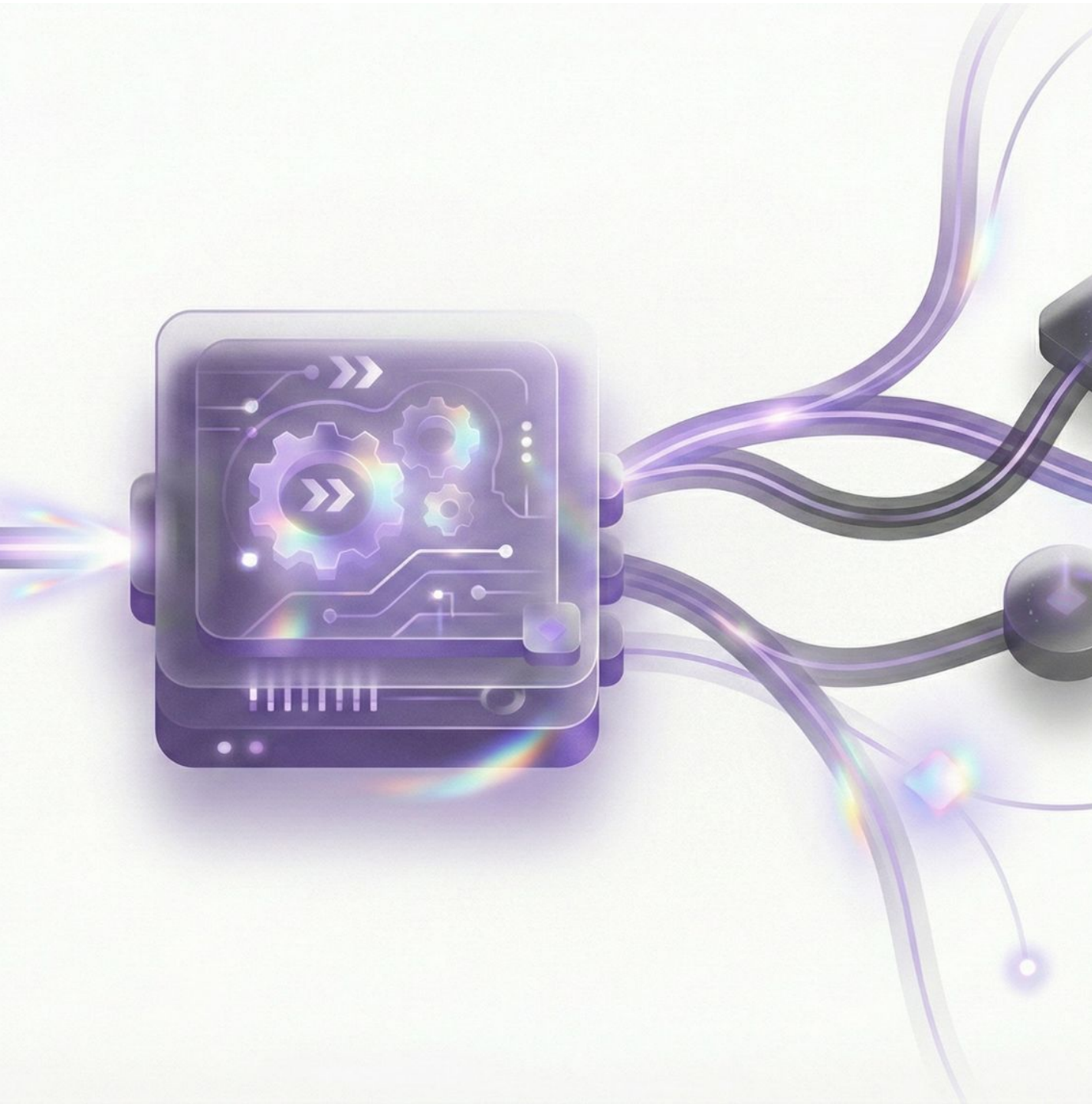


Why This Matters



- Automate complex multi-step workflows requiring multiple tools
- Enhance efficiency and productivity across business operations
- Fosters operational scalability
- Reduce human cognitive load
- Enable 24/7 autonomous operation

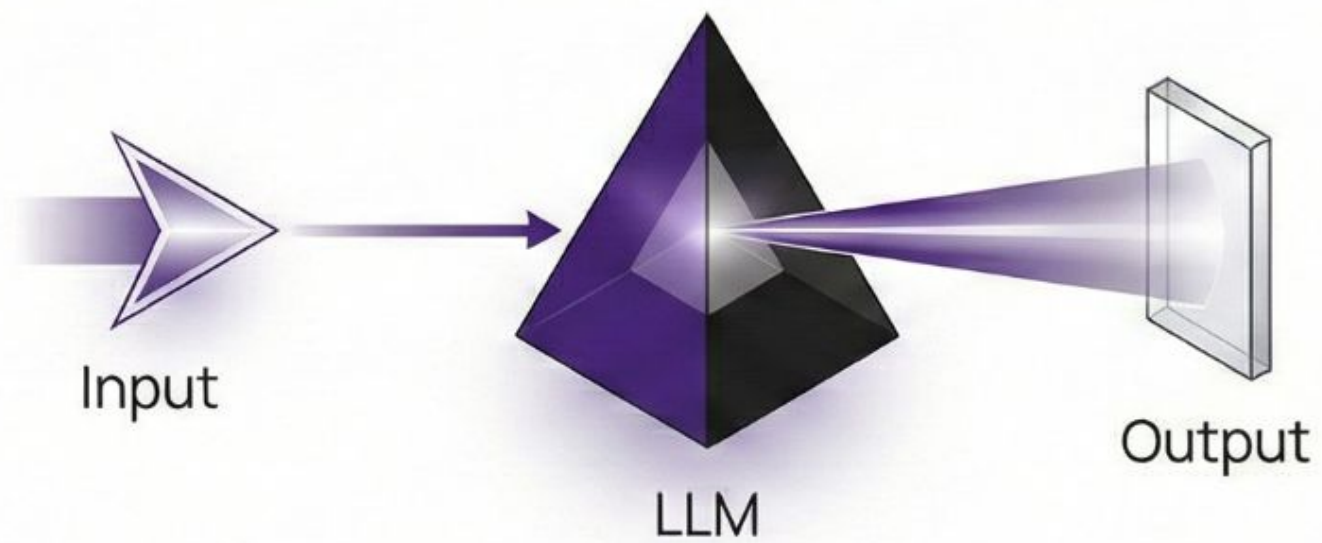
How it Works



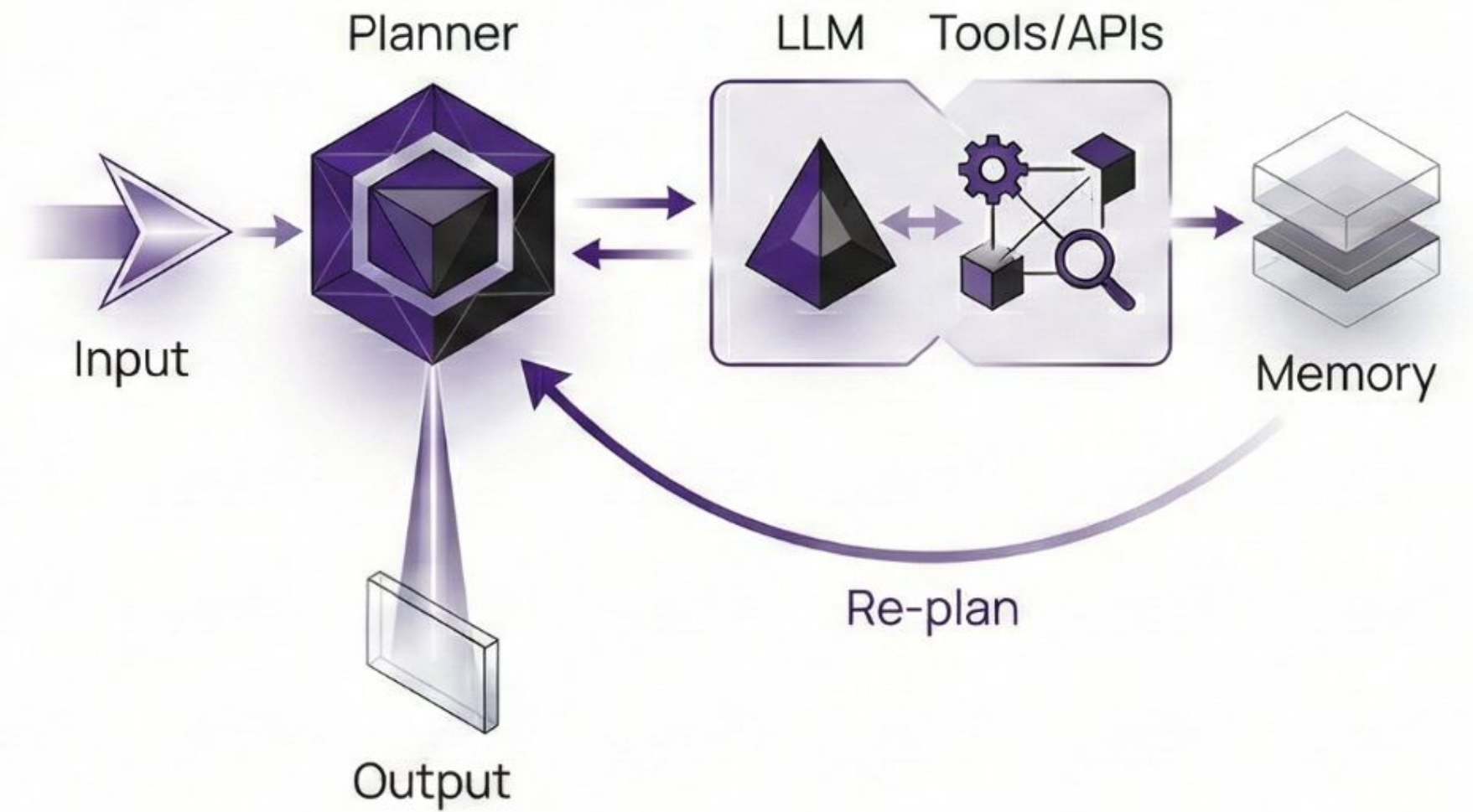
- Autonomy: Autonomously pursue specified goals & objectives
- Planning & Reflection: Decompose goals into subtasks, evaluate progress, and revise plans dynamically
- Tool Use: Access and leverage multiple external tools, APIs, MCPs, A2As, etc.
- Persistent Memory: Maintain context across multi-step processes
- Feedback Loops: Self-correct based on outcomes and observations

AI Chatbots vs AI Agents

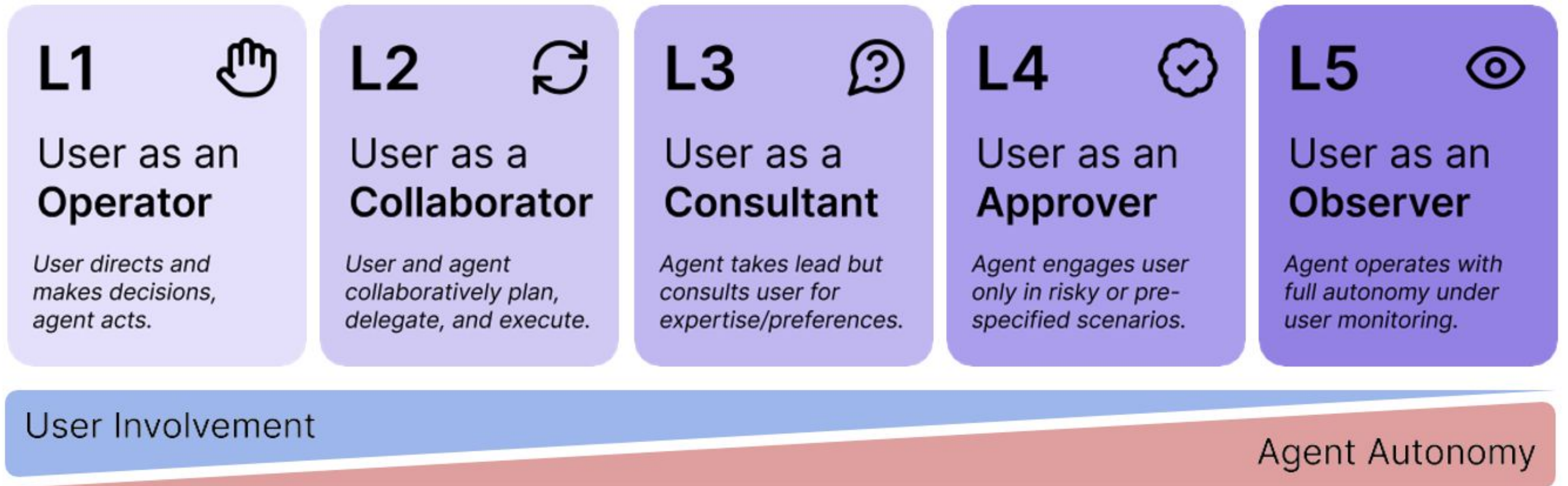
AI Chatbot



AI Agent



Levels of Autonomy



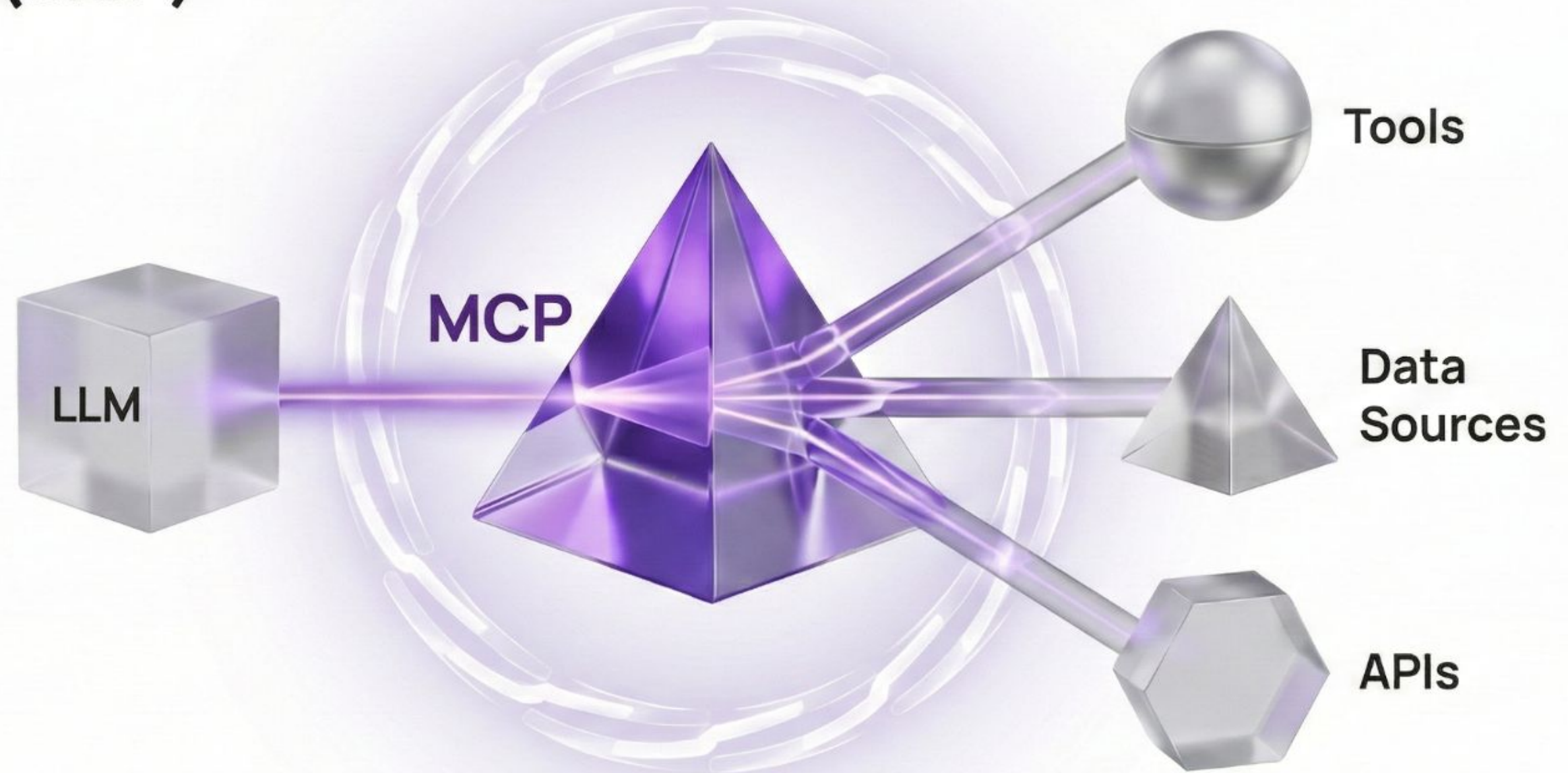
Agentic Protocols

Model Context Protocol (MCP)

Standardized interface:
for connecting LLMs to
tools and data sources.

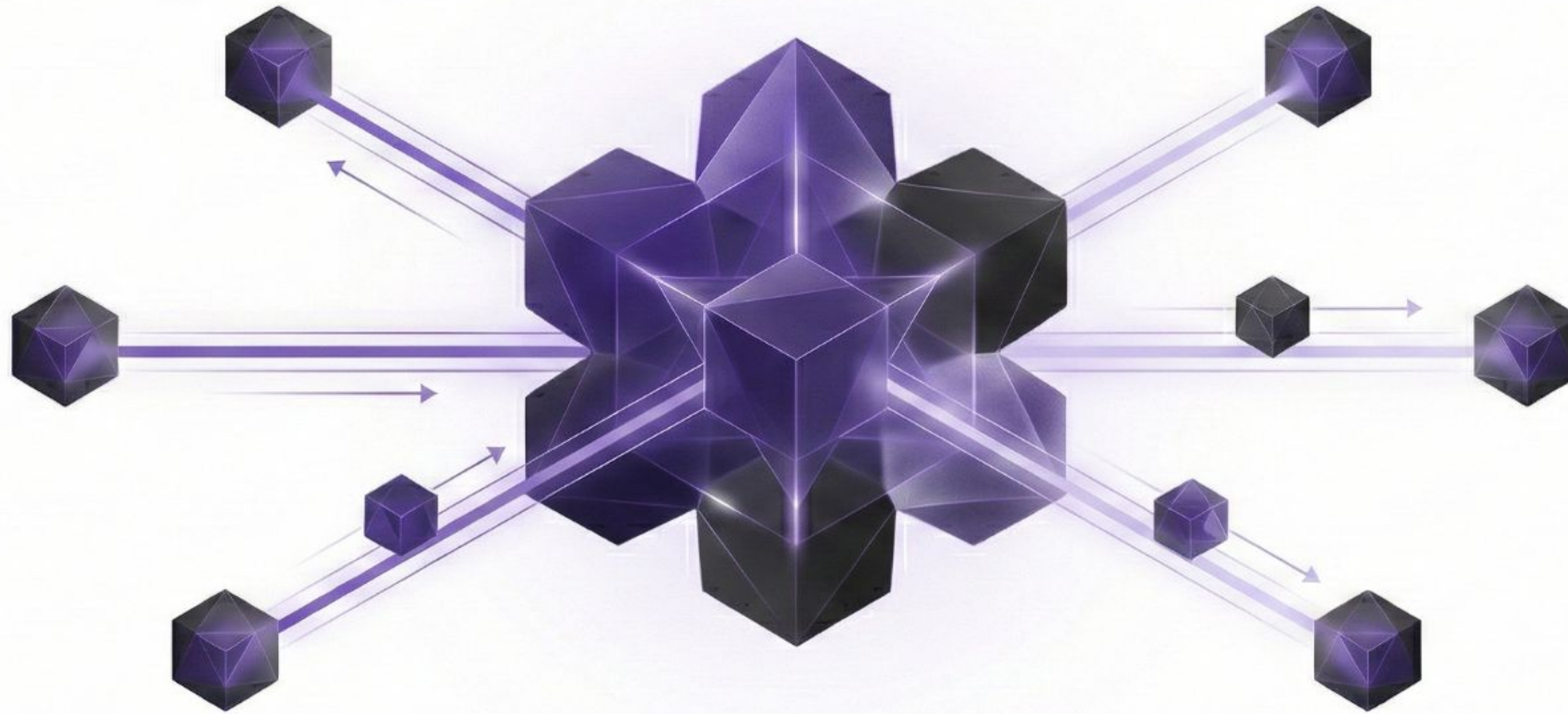
Secure boundaries:
controls what agents can
access and do.

Interoperability: swap
models or tools without
rewriting integrations.



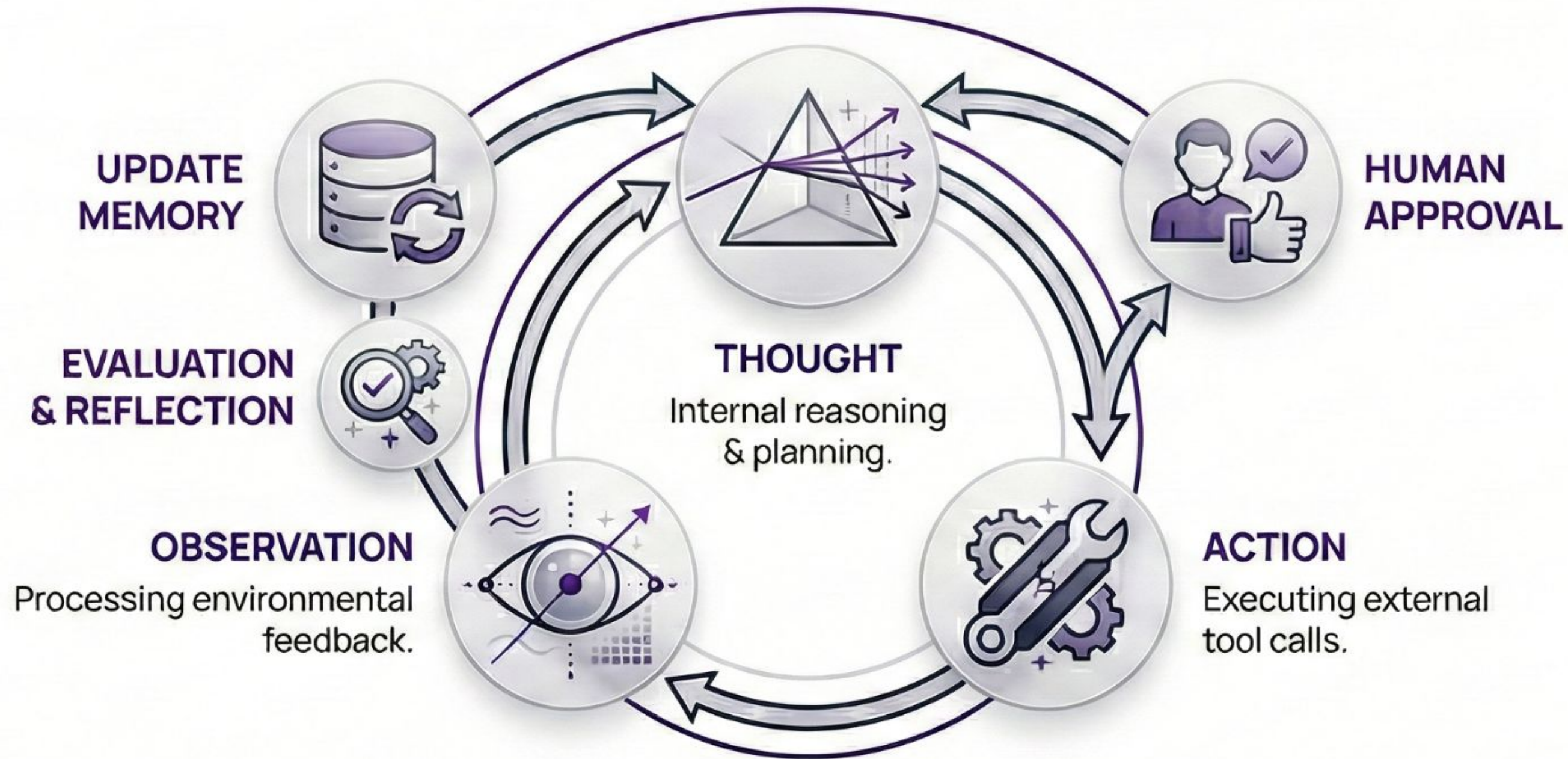
AGENTIC PROTOCOLS

Agent2Agent (A2A)



- **Agent-to-agent communication:** enables agents to discover and interact with each other
- **Delegation:** agents assign subtasks to specialized peer agents
- Supports **multi-agent systems** across organizations and platforms

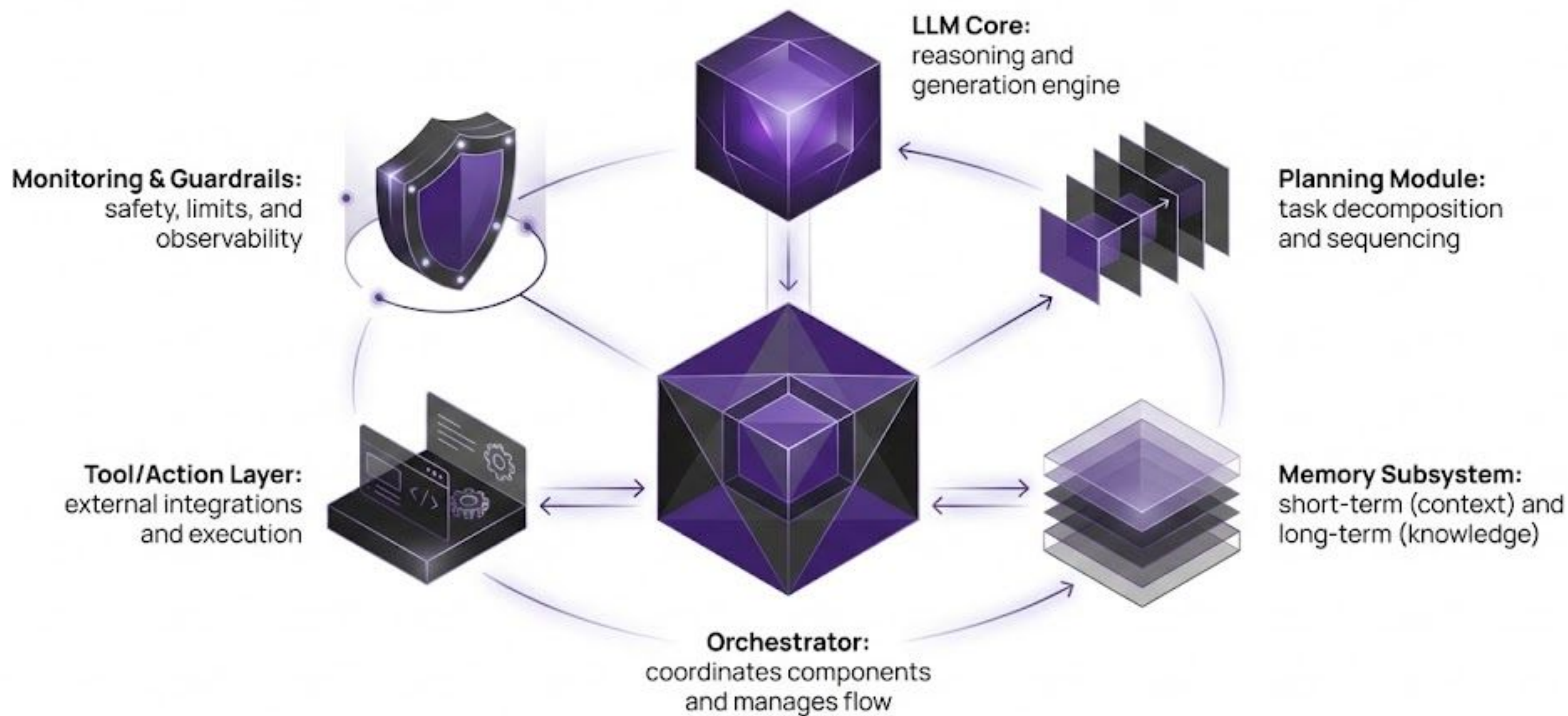
The Extended ReAct Loop



REASONING TRACE

Thought: Need data. **Action:** Query API. **Observation:** Success.
Evaluation: Outcome meets criteria. **Memory:** Updated knowledge base. **Approval:** Human confirmed.

Agent System Architecture



Activity

Architect the Agent

Use Case: A corporate AI travel agent that plans and books multi-city business trips — including flights, hotels, and meeting rooms — while following company travel policy and staying within budget.

Goal: Design the agentic architecture and trace through the Extended ReAct Loop for a real request.

*** Don't use any devices for assistance

*** Follow guidance on activity steps

Activity

Design Your Architecture

The Request: Book a 3-day business trip: New York → Austin → San Francisco. I need flights, hotels near our offices, and a meeting room in each city. Budget: \$5,000. Must comply with company travel policy.

Discuss with your group:

- What architecture components does this agent need and why?
- Walk through the Extended ReAct Loop for the agent's first action
- Where are guardrails and human approval needed?

*** Don't use any devices for assistance

*** Follow guidance on activity steps

Activity

Phase 2 – Curveball

The Curveball: The NYC hotel near your office is fully booked. The next closest option adds \$800 – pushing the trip \$300 over budget.

Discuss with your group:

- How does the agent detect and respond to this failure?
- What does the Extended ReAct Loop look like now?
- Does this require human escalation – why or why not?

*** Don't use any devices for assistance

*** Follow guidance on activity steps

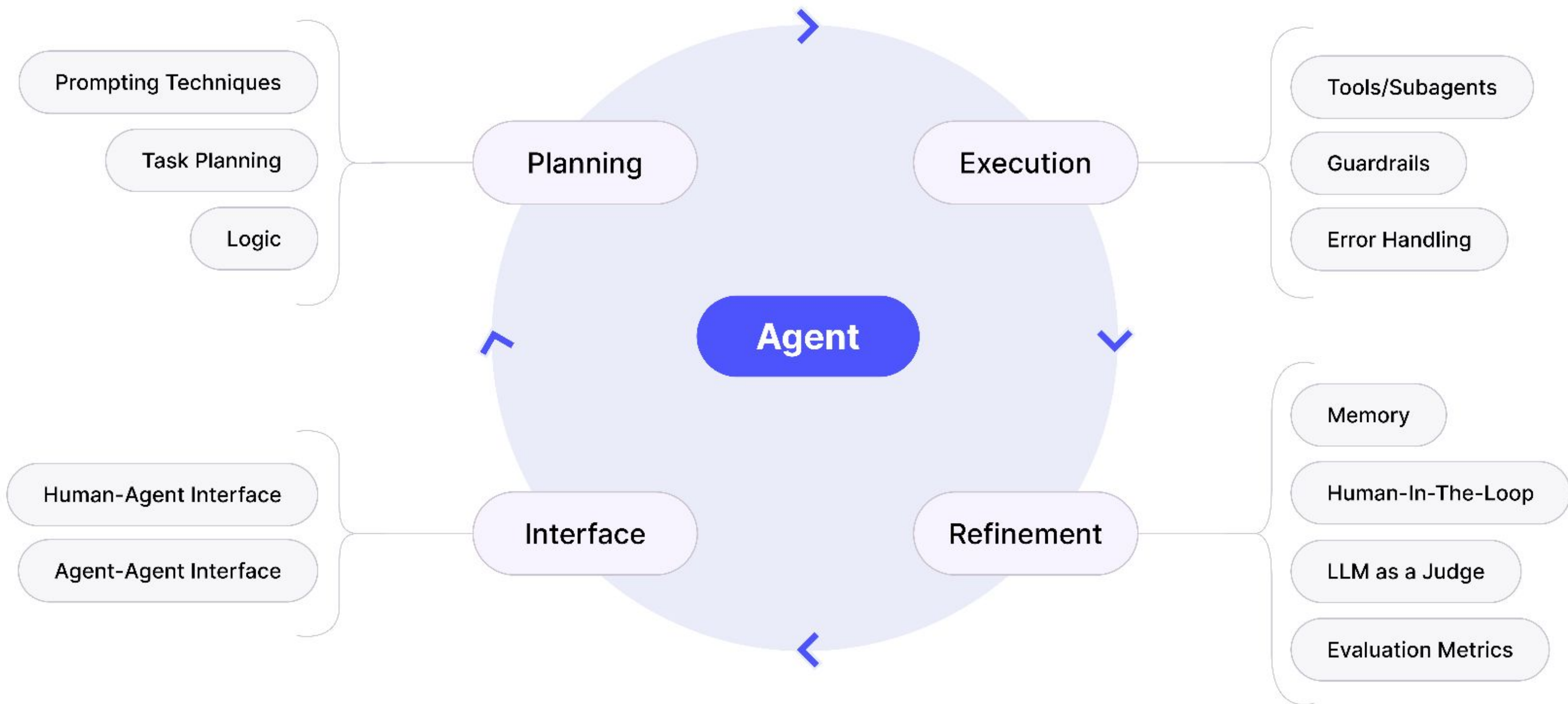
Activity

Discussion:

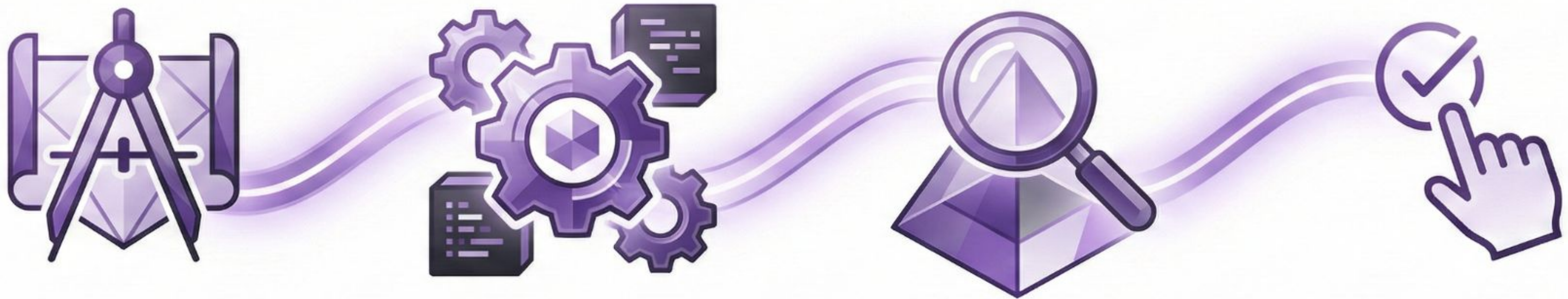
- How did your architecture handle the curveball?
- Which component were most critical during failure recovery?
- Where did you draw the line between agent autonomy and human decisions?
- How would you measure whether this agent is successful?

*** Don't use any devices for assistance

*** Follow guidance on activity steps



Agentic Workflow Orchestration



Planner Agent

defines steps +
assigns work

Execution Agent

performs API calls +
updates state

Reviewer Agent

validates output
(quality + safety checks)

Human Approval

final action
(when risk is high)

Agentic AI in the Real World

- **Customer Support:** agents that resolve tickets end-to-end (triage → lookup → respond → escalate)
- **Software Engineering:** coding agents that plan, write, test, and debug across files
- **Finance:** agents that monitor markets, flag anomalies, and draft analyst briefs
- **Research:** deep research agents that search, synthesize, and compile reports autonomously



Workflow Complexity Spectrum



SIMPLE

single-step, no integrations

Example: summarize a document

MODERATE

multi-step, single-system

Example: research → draft → edit
within one tool

COMPLEX

multi-system, tools orchestration

Example: pull CRM data → generate report →
email stakeholders → log in project tracker

Designing Agentic Workflows

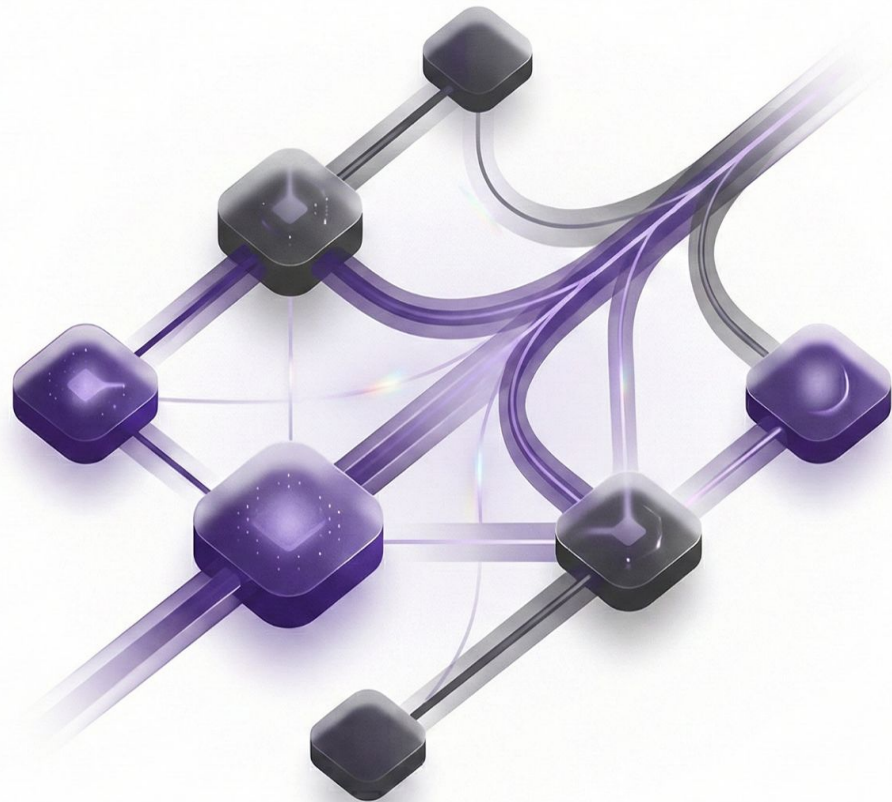
- Value & ROI: does agent complexity justify the gain?
- Feasibility: can the task be decomposed and automated?
- Risk & Guardrails: what can go wrong, and what limits are in place?
- Human oversight: where must a human approve, review, or intervene?
- Success metrics: how will you measure accuracy/trust, efficiency, and safety?

Orchestration Challenges



- Tool selection & safety (prompt injection, permissions)
- State management across steps (context, memory, handoffs)
- Error handling & recovery (timeouts, retries, partial failures)
- Observability (logs, traces, evaluations)
- Fallback paths (degrade gracefully to rules or humans)

Agentic AI Tools & Builders



- OpenAI Agent Builder
- Langflow
- Flowise
- Dify
- N8n Agents
- Make.com Agents

*** Non-exhaustive

Rule of Thumb



Only use agents when your workflow is non-deterministic, needs extra flexibility, or requires adaptive decision-making.

Otherwise, keep it simple.

Activity

YOLO Marketing Spend Agent

Scenario: A company is building an AI agent that optimizes and allocates digital marketing spend. It reads performance data, decides changes, and directly executes campaign budget and bid changes across Google, Meta, and LinkedIn on a regular basis.

The agent is allowed to decide the daily campaign spend amount and can increase or reallocate spend without a fixed cap. Engineers want to build using “dangerously-skip-permissions,” with no scoped access, no approvals, and other YOLO configuration/settings because “AI is so good now.”

If things go wrong, costs could escalate rapidly through both marketing spend and the agent’s own operating costs.

*** Don’t use any devices for assistance
*** Follow guidance on activity steps

YOLO Marketing Spend Agent

Questions:

- What risks do you see in this scenario? Consider costs, model behavior, agent autonomy, tool access and actions, and security and privacy.
- Which risk is most dangerous? What is the worst plausible 72-hour outcome, and what makes the damage compound rather than stay contained?
- Which risks are worth taking for the value, and which are non-negotiable? What principle are you using to decide?
- Where do you draw the autonomy line? What can the agent do without approval, what requires human approval, and what must be blocked?
- What is the minimum viable safe version to launch first? What controls and evidence would you require before expanding autonomy over time?

Activity

*** Don't use any devices for assistance
*** Follow guidance on activity steps